BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING (RAPC-202)

BTECH .3rd SEMESTER | EXAM: NOV /DEC 2025

MAX. MARKS -50 | MAX.TIME – 3 Hr

Note: Attempt any five questions.

Q1. [a] With a neat sketch briefly explain, how an alternating voltage is produced when a coil is rotated in a magnetic field. [3]

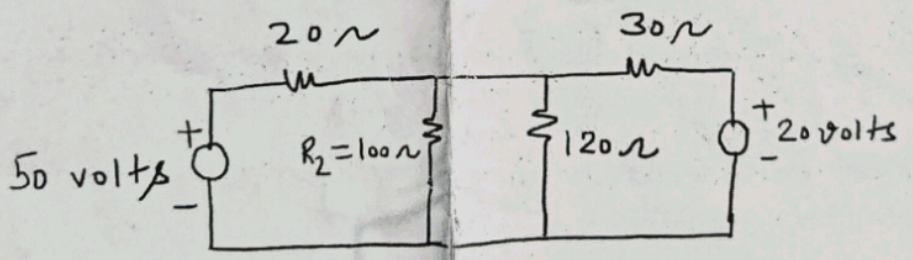
[b] Differentiate between Form factor and Peak factor. Draw the power triangle by clearly mentioning the Active, Reactive and Apparent power. [3]

[c] An alternating current is represented by $i = 400 \sin (157\, t + \pi/6)$ Determine (i) Peak value (ii) Average value (iii) RMS value (iv) Frequency (v) Time period (vi) Form factor (vii) Peak factor (viii) Phase angle of the current. [4]

Q2. [a] A non-inductive resistance of 10 Ω is connected in series with an inductive coil across a 200 V, 50Hz supply. The current drawn by the series combination is 10A. The resistance of the coil is 2 Ω. Determine the (i) Inductance of coil (ii) Power factor (iii) Voltage across coil. [3]

[b] Derive the relationship between the line voltage and phase voltage in 3-phase star connected load. Draw the phasor diagram also. [3]

[c] Using the nodal analysis, find the current through resistance R_2. Use matrix method. [3]



[d] Define speed regulation in dc motor. [1]

Q3. [a] Define armature core and commutator in dc generator. [2]

[b] A Shunt generator supplies 96 A at a terminal voltage of 200 V. The armature and shunt field resistance are 0.1 Ω and 50 Ω respectively. The iron and friction losses are 2500 W. Find the (i) E.M.F generated (ii) copper loss (iii) commercials efficiency [5]

[c] Define and derive the expression of armature torque of a DC motor and show that it is proportional to flux per pole and armature current. [3]

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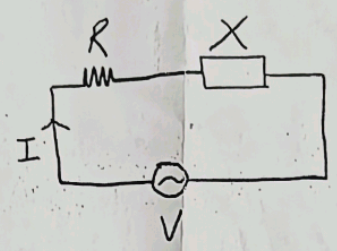
Q4. [a] Explain the production of rotating field in 3- phase induction motor and show that the resultant flux is of constant value =3/2 and at which speed this resultant flux is rotated around the stator. [4]

[b] A 10 KVA, 2000/400 V single phase transformer has R1 = 5Ω, X1 = 12 Ω, R2 = 0.2Ω, X2 =0.48 Ω. Determine the equivalent impedance of the transformer referred to primary side. [3]

[c] In the circuit shown below the applied voltage is given by (0+ j10) V and current is (0.8 + j 0.6) A. Determine the value of R and X and also indicate if X is inductive or capacitive. [3]



Q5. [a]Differentiate between Ammeter and Voltmeter. Explain why power in AC circuits cannot be measured by using ammeter and voltmeter. [2]

[b] Explain the essentials of an indicating instrument. [3]

[c] Explain the principle, construction and working of PMMC type of instruments. [5]

Q6. [a] Define FET and differentiate between a BJT and FET. Name any two types of BJT and FET. [3]

[b] What is a semiconductor and differentiate between intrinsic and extrinsic semiconductor. Give any two examples of semiconductors. [3]

[c] A particular number is written as 132 in radix-4 representation. Find the equivalent of this number in radix-5 representation [2]

[d] If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, Find the decimal value of x + y [2]